LiteYOLO-ID: A Lightweight Object Detection Network for Insulator Defect Detection

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***Abstract*— Insulator defect detection is of great significance to ensure the normal operation of power transmission and distribution networks. In response to the problems of low speed, low accuracy, and difficulty in deploying to embedded terminals in existing insulator defect detection, this article proposes a lightweight insulator defect detection model based on an improved YOLOv5s, named LiteYOLO-ID. First, to significantly reduce the model parameters while maintaining detection accuracy, we design a new lightweight convolution module called ECA-GhostNet-C2f (EGC). Second, based on the EGC module, we construct the EGC-CSPGhostNet backbone network, which optimizes the feature extraction process and achieves model compression. Additionally, we design a lightweight neck network, EGC-PANet, to further reduce the parameter count and achieve efficient feature fusion. Experimental results show that on the IDID-Plus dataset, compared to the original YOLOv5s model, not only does LiteYOLO-ID reduce the model parameters by 47.13%, but it also improves the mAP (0.5) by 1%. Furthermore, the generalization of the model is validated on the Pascal VOC dataset and the SFID dataset. Importantly, after TensorRT optimization, the inference speed of the LiteYOLO-ID algorithm on the Jetson TX2 NX reaches 20.2 frames/s, meeting the real-time detection requirements of insulator defects. Our code, weight models, and datasets can be obtained at the following URL: https://github.com/LuYang-2023/Insulator-defect-detection.**

***Index Terms*— Deep learning, insulator defect detection, lightweight, quantification and deployment, YOLO.**

I. Introduction

As SOCIETAL, economic, and technological advancements continue, the scale of transmission lines consistently expands. Due to prolonged exposure to harsh environments, insulators are prone to faults, and pose obstacles to the stable operation of the power grid. Consequently, insulator defect detection has become an indispensable and crucial component in the detection of faults in transmission lines [1].

In the past decade, with the substantial development of deep learning technology and computer hardware [2], [3], [4], [5], UAVs have opened new avenues for the accurate identification and localization of defective insulators [6].

Detecting insulator defects using drones requires highly maneuverable capabilities. Therefore, edge computing devices are typically chosen as deployment platforms. However, due to the computational limitations of edge devices, the inference execution time of the object detection model on edge computing devices is considerably long, and a substantial amount of storage space is required. Hence, one urgent issue to address is designing a model suitable for edge computing devices that can perform real-time and accurate insulator defect detection.

To address the aforementioned issues, this article presents a lightweight detection model, LiteYOLO-ID, tailored to meet the practical application demands of detecting insulator defects using embedded devices like drones. First, to reduce the model parameter count, we introduce a lightweight convolution module, ECA-GhostNet-C2f (EGC), drawing inspiration from the design principles of GhostNet [7] and C2f modules, and enhancing the capture capability of crucial information of detection targets through the ECA attention mechanism [8]. Second, building upon the EGC module, we devise a lightweight backbone network, EGC-CSPGhostNet, to decrease the parameter quantity of the model. Additionally, to optimize model parameters and enhance feature fusion efficiency, we propose a lightweight neck network, EGC-PANet. To address the issue of singular insulator defect types, we curate and annotate a vast array of insulator defect images, constructing the IDID-Plus dataset using fogging data augmentation techniques. Finally, we employ TensorRT [9] to quantize the refined model and deploy it on Jetson TX2 NX, thereby validating the model's effectiveness in real-world deployment scenarios.

The main contributions of this article are as follows.

1) We propose LiteYOLO-ID, a lightweight insulator defect detection model specifically designed for embedded devices, effectively balancing the model parameter count and detection accuracy.
2) We design a lightweight convolution module, EGC, and construct a lightweight backbone network, EGC-CSPGhostNet, based on this module to reduce the model parameter count and enhance feature extraction. Additionally, we build a lightweight neck network, EGC-PANet, to further reduce the model parameters and accelerate the feature fusion process.
3) We curate a large collection of insulator defect images, annotate them, and create the IDID-Plus dataset by applying fogging augmentation techniques.

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4) We utilize TensorRT to quantize LiteYOLO-ID with 16-bit floating-point precision and deploy it on the edge computing platform Jetson TX2 NX to verify the actual detection efficiency of the model.

The remaining structure of this article is as follows. Section II reviews the relevant literature on insulator defect detection and provides an overview of deploying object detection models on edge platforms. Section III will provide a detailed introduction to the LiteYOLO-ID model and its corresponding modules. Section IV will cover experimental details such as dataset preparation, parameter settings, and evaluation metrics. The effectiveness of the proposed method will be demonstrated through experiments, and a thorough analysis of the experimental results will be presented. Finally, Section V will conclude the article.

II. Related Work

In this section, we review the relevant literature in the current field of insulator defect detection and provide an overview of research related to deploying lightweight object detection models on embedded devices.

A. Insulator Defect Detection Overview

The earliest method for detecting insulator defects was manual inspection. However, manual inspection is labor-intensive, and workers sometimes have to climb mountains and traverse rugged terrain. They also endure extreme weather conditions, facing both severe cold and scorching heat.

In the early stages, insulator defect detection was generally based on traditional image processing techniques. Zhao et al. [10] proposed an insulator positioning method based on directional angle and binary shape prior knowledge to meet the requirements of insulator positioning. Liao and An [11] introduced an approach that combines the local features of insulator images with the spatial order of the images, achieving the detection of overhead insulators.

In recent years, with the breakthroughs in deep learning theory, convolutional neural networks (CNNs) have significantly improved the accuracy of image recognition, leading to the pervasive integration of deep learning techniques across various fields. Currently, object detection algorithms are primarily divided into two categories based on their detection approach. One category is two-stage object detection algorithms, including R-CNN [12], Fast R-CNN [13], Faster R-CNN [14], and Mask R-CNN [15]. Zhao et al. [16] improved Faster-RCNN with feature pyramid network (FPN) and applied an adaptive threshold algorithm in the HSV color space for image segmentation. This method achieved an mean average precision (mAP) of 90.8% on glass insulators and 91.7% on composite insulators. Zhai et al. [17] proposed a CNN based on mixed knowledge domains (HK R-CNN) to address the issues of insufficient samples and small targets, achieving a detection average precision (AP) of 79.27%. Dian et al. [18] proposed an insulator defect detection method named Faster R-Transformer based on the attention mechanism of Transformer and Faster R-CNN, achieving an accuracy of 99.78%. Chen et al. [19] combined EfficientNet and DWConv to construct a lightweight detection model called Faster R-CNN-tiny. This model not only improves defect detection in complex environments but also highlights the potential of deploying lightweight models on edge devices.

Another category is the one-stage object detection algorithms, with the most representative ones being the YOLO series [20], [21], [22], [23], [24], [25], [26], [27], [28], SSD series [29], [30], RetinaNet [31], and so on. Zhang et al. [32] constructed a dataset of 13 000 insulator defect images to address the issue of insufficient samples, and they introduced an attention mechanism to improve YOLOv5, achieving a detection accuracy of 99.5%. Zhang et al. [33] proposed a densely connected FPN (DC-FPN) based on YOLOv3, which reduces the parameter count of the insulator defect detection model and improves detection accuracy. Song et al. [34] proposed an improved version of Deformable YOLOX based on YOLOX, which enhances the accuracy of detecting dense small targets in insulator detection. Hao et al. [35] introduced the CSP-ResNeSt backbone network structure and incorporated Bi-SimAM-FPN to enhance YOLOv4. Their insulator detection accuracy reached 95.63%. Qi and Ye [36] combined GhostConv, inverted residual structure, and CARAFE upsampling operator to improve YOLOv5, reducing computational complexity while achieving a detection accuracy of 93.6%. Xie et al. [37] improved the YOLOv4-tiny model by incorporating transfer learning and SE attention mechanism, and adding small object detection layers to enhance the recognition capability of small objects, achieving an AP of 91.45% in insulator detection. Wang et al. [38] reduced the model parameter count by adopting DWConv and an improved feature fusion module based on YOLOv5, and optimized performance using knowledge distillation, achieving a detection accuracy of 82.8% in insulator defect detection. Ding et al. [39] improved YOLOv5s by integrating GhostConv and CA attention mechanism, and introduced EVCBlock to enhance the neck network for better feature representation, achieving a detection accuracy of 94.2% in insulator defect detection. Bao et al. [40] optimized the YOLOv5 detection model by introducing CA attention mechanism and BiFPN to better handle small-scale and diversified features, achieving a detection accuracy of 89.1%. Zhang et al. [41] integrated Ghost module and CBAM attention mechanism into the YOLOv5 architecture to create a model that saves computational resources while maintaining detection accuracy and speed.

Although two-stage detection models exhibit excellent performance in terms of accuracy, their large parameter count makes deployment on embedded hardware challenging. On the other hand, while one-stage detection models are suitable for embedded devices, their detection accuracy is relatively low, failing to meet the requirements of practical transmission line inspection. Therefore, this article proposes a lightweight detection model, LiteYOLO-ID, to meet the demands of practical transmission line inspection.

B. Object Detection for Edge Platforms

In practical application scenarios, target detection models must be deployed on edge computing devices. However,

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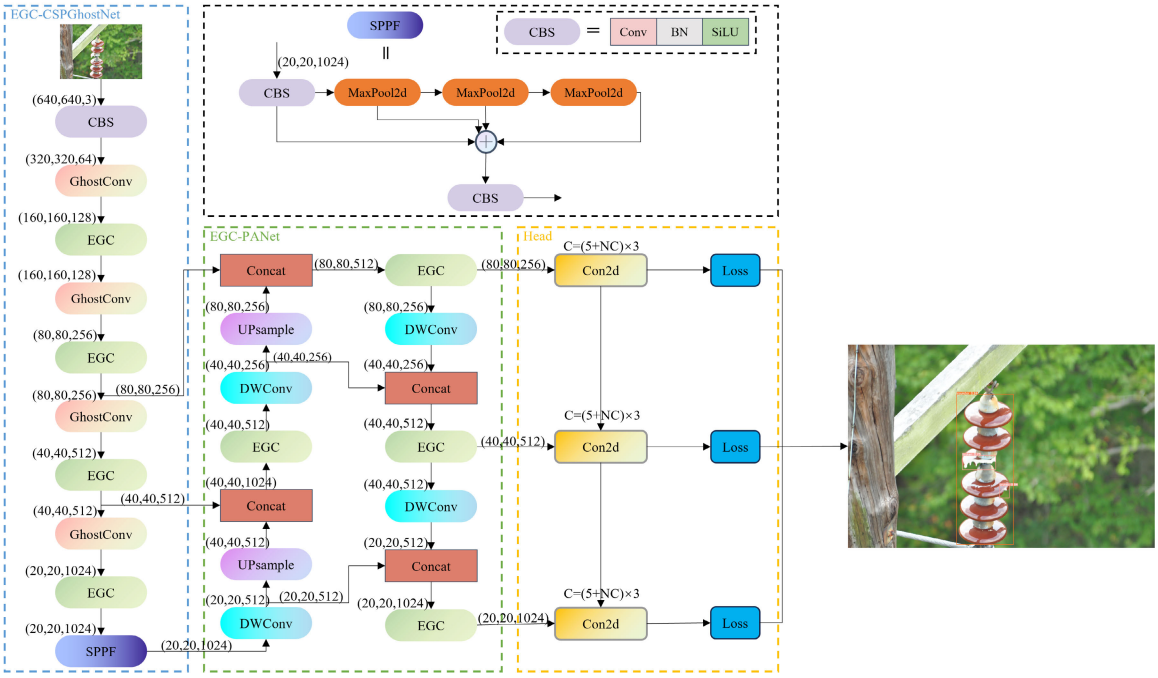


Fig. 1. Overall structural diagram of LiteYOLO-ID.

considering the limited computational power of edge computing devices, the quantized deployment of target detection networks becomes a crucial factor.

Liu et al. [42] improved the YOLOv5 network and proposed RepYOLO. By optimizing the model using C++ combined with TensorRT on Jetson Xavier NX, they successfully achieved intelligent power inspection. Yue et al. [43] proposed an ultra-lightweight detection model named YOLO-GS and optimized the model using TensorRT. The deployment on Jetson Xavier NX demonstrated excellent performance, achieving a detection accuracy of 99.38% for empty bowls. Liu et al. [44] proposed a lightweight robot object detection algorithm based on an improved version of YOLOv5, aiming to enhance model detection accuracy and reduce computation time. The model's detection performance was validated on the NVIDIA Jetson TX2 platform. Therefore, this article deploys LiteYOLO-ID on Jetson TX2 NX to verify its detection effectiveness in practical engineering applications.

III. Methods

This section first introduces the backbone network of LiteYOLO-ID and its related components. It then provides a detailed explanation of its neck and head network architectures. Finally, it delves into the implementation of TensorRT acceleration. Fig. 1 shows the overall structural diagram of LiteYOLO-ID.

A. EGC-CSPGhostNet Backbone Network

The backbone network of LiteYOLO-ID is EGC-CSPGhostNet, primarily comprising four modules: CBS, GhostConv, EGC, and SPPF.

1) CBS Module: The CBS module consists of ordinary convolution, Batch Normalization, and SiLU activation function. It is used to extract better features from the input image.

2) GhostConv Module: Although traditional convolutional modules are good at feature fusion and transmitting contextual information between feature maps, they have a large parameter count and contain many redundancies. To reduce the model's parameter size, the GhostConv module is introduced into the backbone network. The GhostConv module first obtains the intrinsic feature map using a regular convolution. Subsequently, it generates Ghost feature maps for each channel feature map of the intrinsic feature map using cheap operation Φ. Finally, the initially obtained intrinsic feature map is concatenated with the Ghost feature map to produce the ultimate result, denoted as Output. The operational principles are illustrated in Fig. 2.

3) EGC Module: To obtain richer gradient flow information and improve model performance while reducing parameter count, we drew inspiration from the C2f module and GhostNet concepts. We adopted the GhostBottleneck to replace the bottleneck structure in the C2f module, thereby reducing the module's parameter count. The architecture of the GhostBottleneck is shown in Fig. 3. However, this modification led to a decrease in accuracy. To optimize the detection accuracy of the model, we introduced the ECA attention mechanism at the beginning of the module, thus creating the EGC module. This module ensures an improvement in detection accuracy while reducing the parameter count. The structure of the EGC module is illustrated in Fig. 4.

4) SPPF Module: The SPPF module is an improvement on the SPP structure, aimed at reducing the computational load of the model, thereby significantly increasing its speed.

B. EGC-PANet Neck Network

The neck network of LiteYOLO-ID employs EGC-PANet, whose architectural design draws inspiration from the ideas of FPN [45] and PANet [46]. FPN is a method that fuses

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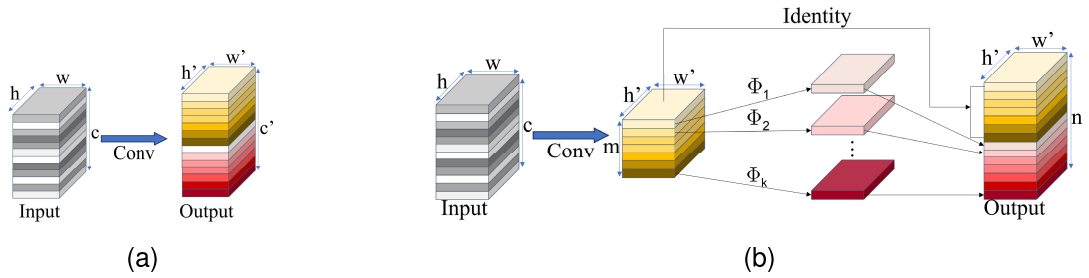


Fig. 2. Operation diagram of the GhostConv module. (a) Output feature map of a conventional convolution. (b) Output feature map of the GhostConv module.

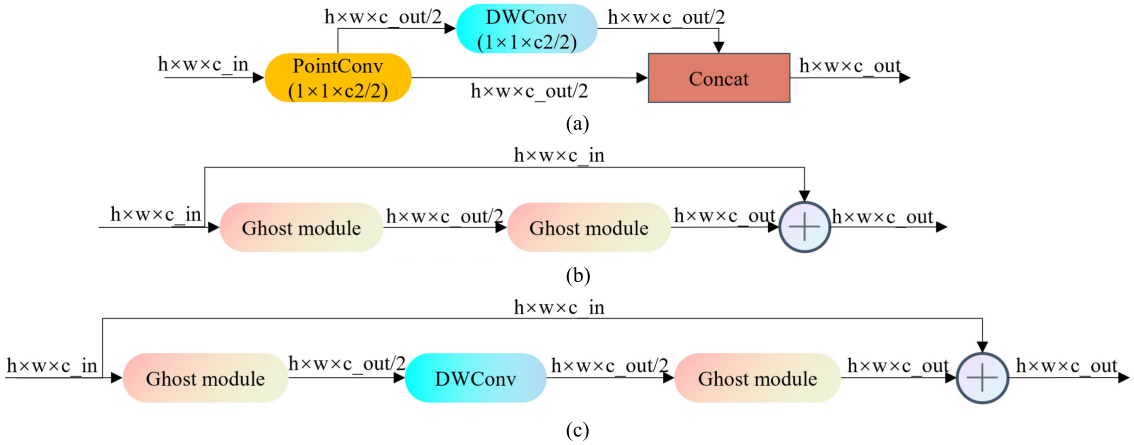


Fig. 3. Structure diagram of GhostBottleneck.

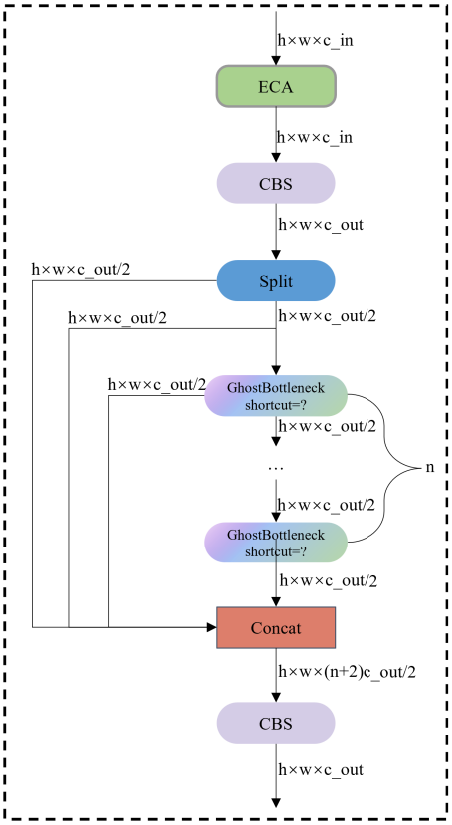


Fig. 4. Structure diagram of the EGC module.

high-level features from top to bottom with low-level features, but after passing through several intermediate layers, the bottom-level information becomes very blurred. Therefore, PANet is introduced to add a bottom-up module, enhancing the network's feature fusion capability. To further improve the model's feature capturing ability, enhance detection accuracy, and reduce parameter count, we incorporated the EGC module. However, even with the EGC module, there is still some redundancy in the neck part. Therefore, we further introduced the DWConv [47], [48] module in the neck part to streamline the model parameters and enhance the model's efficiency and performance.

C. YOLO Head Network

The main function of the head network is to determine the location and category of the detection targets from the extracted feature maps. We use the head network of YOLO, which performs multiscale object detection on the extracted feature maps through three detection heads. It generates detection boxes, classifies these boxes, and conducts regression on each box to produce the final prediction results.

D. TensorRT Acceleration

TensorRT is a high-performance deep-learning inference optimizer provided by NVIDIA, which can offer low-latency and high-throughput inference deployment for deep learning applications. After LiteYOLO-ID is trained on a PC, the trained weight file can be converted into ONNX [49] format. This allows the use of TensorRT for acceleration and inference, thereby enhancing the model's inference speed on edge

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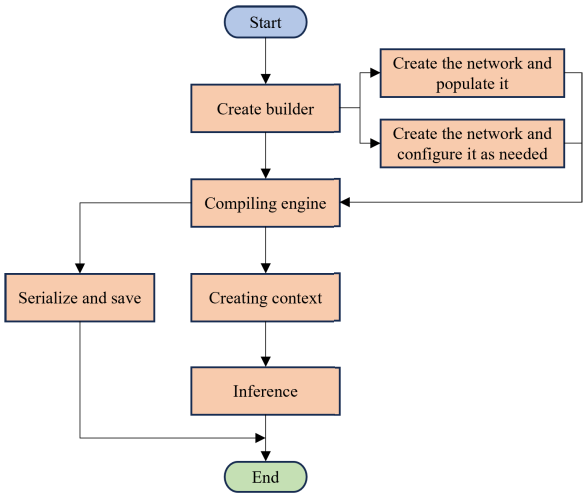


Fig. 5. TensorRT acceleration flowchart.

TABLE I
LABEL INFORMATION FOR THE IDID-PLUS DATASET

Dataset	Labels name	Number of labels
IDID-Plus	Flashover	5030
	Broken	1862
	Insulator	2862

computing devices. The TensorRT acceleration workflow is illustrated in Fig. 5.

TensorRT utilizes layer and tensor fusion technology, combining frequently occurring structures in the network, such as convolutional layers, batch normalization layers, and activation layers, into a single structure to reduce the number of CUDA [50] launches, thus achieving time optimization. During the inference phase, the model's requirement for parameter precision is lower, allowing the use of half-precision floating-point numbers (FP16) to represent model parameters. This lower data precision can reduce memory usage and latency, and also decrease the model size.

IV. EXPERIMENTS

In this section, we first introduce the construction of the IDID-Plus dataset, the experimental environment, and evaluation metrics. Subsequently, comparative experiments and ablation experiments are conducted, followed by an in-depth analysis.

A. Experimental Configuration

1) Dataset: Due to the limited number of insulator defect image samples and the singularity of defect types, this article has reannotated the collected insulator defect images to obtain the IDID-Plus dataset, which features multiple types of defects. The label information of the IDID-Plus dataset is as shown in Table I.

2) Offline Data Augmentation: Due to the actual occurrence of foggy weather during power inspections, this article applies a synthetic fog operation to the insulator dataset based on the atmospheric scattering model [51]. Considering that a single fogging algorithm may not provide sufficient generalization for the dataset, another common fogging algorithm is introduced. This involves generating a fog layer through random noise and blending this fog layer with the original image. Fig. 6 presents a comparison between images with added fog and normal images. Utilizing the aforementioned two synthetic fog algorithms, we randomly selected 50% of the images from the collected insulator defect dataset and subjected them to fogging, thereby obtaining the final IDID-Plus insulator defect dataset.

3) Experimental Settings: This article utilizes two computing devices. One is a PC device employed for training, equipped with an NVIDIA RTX 3060 GPU with 6GB of video memory. The software configuration includes PyTorch version 2.0.1 and CUDA version 12.1. The second device is a Jetson TX2 NX utilized for edge detection, featuring NVIDIA Pascal GPU hardware. The software setup on this device includes PyTorch version 1.8.0 and CUDA version 10.2. Fig. 7 illustrates the entire experimental process.

The training procedure takes place on a personal computer, with the dataset partitioned into a training set comprising 2504 images and a validation set comprising 248 images. We employ the Adam optimizer [52] to fine-tune hyperparameters, establish a batch size of 16, and carry out training for 300 epochs. The model achieving the highest performance during training is then saved.

The testing process is carried out on the Jetson TX2 NX development board. Initially, the trained best model is imported into the development board. Subsequently, the model is quantized using TensorRT with FP16. At last, the transformed model is tested using test images, and the results are saved.

4) Evaluation Metrics: The objective of this article is to achieve model lightweighting while ensuring detection accuracy. We employ various metrics, including parameter count (Parameters), floating-point operations per second (FLOPs), AP, mAP, precision (P), and recall (R), to comprehensively assess the model's performance. The calculation formulas for these metrics are as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \times 100\% \tag{1}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \times 100\% \tag{2}$$

$$\text{AP} = \int_0^1 p(r)dr \tag{3}$$

$$\text{mAP} = \frac{\sum_{i=1}^{N} \text{AP}_i}{N}. \tag{4}$$

In the above formulas, TP represents the number of correctly detected targets, FP represents the number of targets incorrectly detected, and FN denotes the number of missed targets. The symbol $p(r)$ is used to represent the precision-recall curve, where N denotes the number of categories of detected objects. In this experiment, N is set to 3, specifically involving three target categories: normal insulators, broken insulators, and flashovers.

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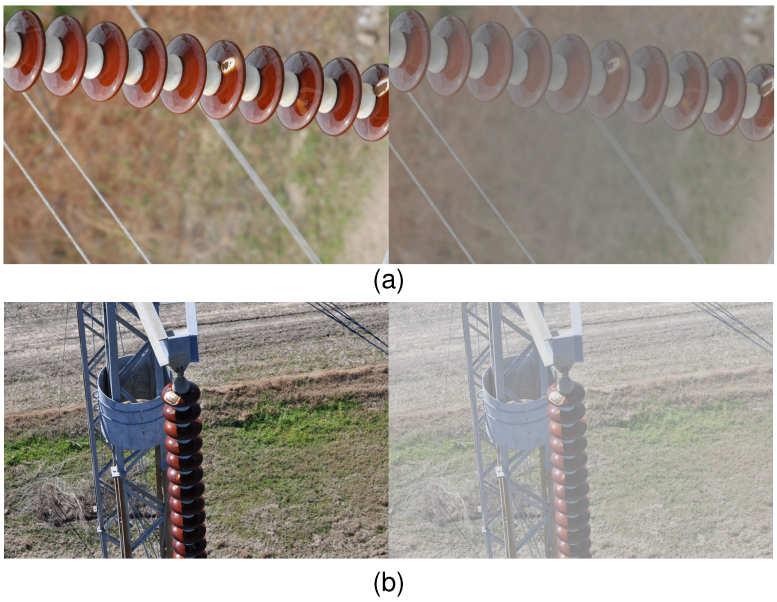


Fig. 6. Comparison of foggy images and normal images. (a) Comparison of images with atmospheric scattering model fogging algorithm and normal images. (b) Comparison of images with ordinary fogging algorithm and normal images.

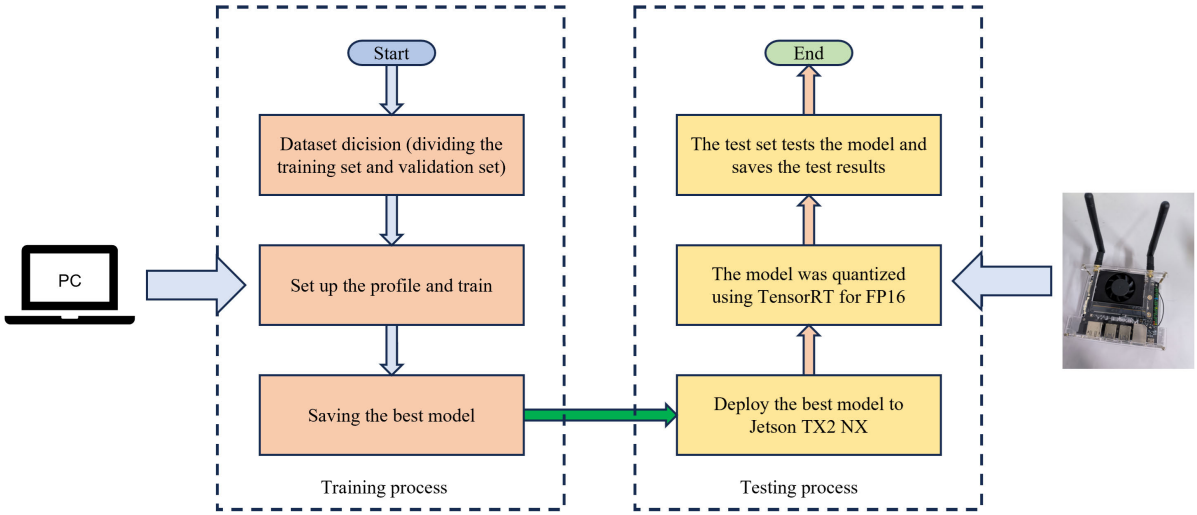


Fig. 7. Experimental process diagram.

B. Experimental Results Comparison

1) Comparing Different Attention Mechanisms: To verify the effectiveness of the ECA attention mechanism in the EGC module, this article employs other mainstream attention mechanisms for comparative experiments. Table II shows the experimental comparison results using different attention modules. On the IDID-Plus dataset, models utilizing the ECA attention mechanism outperform those employing SE [53], CBAM [54], and SimAM [55] attention mechanisms. This discrepancy can be attributed to the ECA attention mechanism's lower parameter count and enhanced network coherence, rendering it more suitable for lightweight network applications.

2) Performance Comparison of Different Lightweight Networks: To validate the model's performance, this article integrates two lightweight networks, MobileNetV3 [56] and ShuffleNetV2 [57], with YOLOv5s, and compares them with LiteYOLO-ID. The experimental results are presented in Table III. While MobileNetV3 and ShuffleNetV2 significantly reduce the model's parameter count and FLOPs, there is a noticeable decrease in accuracy attributed to their limited feature extraction capabilities. In contrast, our proposed algorithm achieves a substantial reduction in parameters and FLOPs without compromising model accuracy.

3) Comparing Performance With Current Mainstream Object Detection Algorithms: To further validate the performance of LiteYOLO-ID, we compared it with mainstream YOLO series detection algorithms. Additionally, we compared it with other detection algorithms that have shown outstanding performance in the field of insulator defect detection. These algorithms include YOLOv5-GCDNet proposed by Qi and Ye [36], improved YOLOv4-tiny proposed by Xie et al. [37], ML-YOLOv5 proposed by Wang et al. [38], Faster R-CNN-tiny proposed by Chen et al. [19], GC-YOLO proposed by Ding et al. [39], BC-YOLO proposed by Bao et al. [40], and the improved YOLOv5 proposed by

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TABLE II
EXPERIMENTAL RESULTS COMPARING DIFFERENT ATTENTION MECHANISMS

Dataset	Model	Param.(M)	FLOPs(G)	P	R	mAP(0.5)	mAP(0.5-0.95)
IDID-Plus	+SE [53]	3.77	8.8	76.2%	60.5%	63.9%	38.4%
	+CBAM [54]	3.78	8.8	74.7%	59.8%	62.9%	36.1%
	+SimAM [55]	3.71	8.7	79.2%	59.1%	63.8%	38.4%
	+ECA(ous)	3.71	8.7	83%	59.5%	65.1%	37.5%

TABLE III
EXPERIMENTAL RESULTS COMPARING DIFFERENT LIGHTWEIGHT NETWORK CONFIGURATIONS

Dataset	Model	Param.(M)	FLOPs(G)	P	R	mAP(0.5)	mAP(0.5-0.95)
IDID-Plus	YOLOv5s [24]	7.02	15.8	75.5%	60.9%	64.1%	39.4%
	YOLOv5s-MobileNetv3 [56]	3.54	6.3	73.1%	54%	58.4%	34.9%
	YOLOv5s-ShuffleNetv2 [57]	3.79	8.0	68.5%	57.9%	58.6%	34.8%
	LiteYOLO-ID(ours)	3.71	8.7	83%	59.5%	65.1%	37.5%

TABLE IV
EXPERIMENTAL RESULTS FOR LIGHTWEIGHT OBJECT DETECTION ALGORITHM COMPARISON

Dataset	Model	Param.(M)	FLOPs(G)	P	R	mAP(0.5)	mAP(0.5-0.95)
IDID-Plus	YOLOv5n [24]	1.76	4.1	71.1%	59.4%	60.7%	36.7%
	YOLOv7-tiny [27]	6.02	13.2	77.6%	63.6%	64.9%	38.5%
	YOLOv8s [28]	11.13	28.4	81.1%	62.9%	66.7%	42.7%
	YOLOv5-GCDNet [36]	3.75	8.8	75.4%	59.9%	62.9%	36.7%
	Improved YOLOv4-tiny [37]	23.33	58.8	80.3%	65.3%	68%	41.1%
	ML-YOLOv5 [38]	3.73	8.9	80.7%	58.7%	63.7%	38.6%
	Faster R-CNN-tiny [19]	10.65	22.9	78.5%	58.3%	63.6%	39.1%
	GC-YOLO [39]	9.64	29	73%	61%	61.7%	38.7%
	BC-YOLO [40]	7.25	16.5	81.2%	60.9%	65.3%	40.3%
	Improved YOLOv5 [41]	3.8	9.7	74.7%	57.3%	61.5%	38%
	YOLOv5s [24]	7.02	15.8	75.5%	60.9%	64.1%	39.4%
	LiteYOLO-ID(ours)	3.71	8.7	83%	59.5%	65.1%	37.5%

Zhang et al. [41]. We strictly reproduced these models according to the model specifications and experimental settings in the original literature, ensuring training and evaluation of all models under the same dataset and experimental conditions to guarantee the fairness of the comparison.

The experimental results are shown in Table IV. Although LiteYOLO-ID has more parameters and FLOPs than YOLOv5n, its detection performance is superior to YOLOv5n. While YOLOv8s and Improved YOLOv4-tiny have better detection performance than LiteYOLO-ID, their parameter count and FLOPs are an order of magnitude higher. LiteYOLO-ID's accuracy is slightly lower than BC-YOLO, but its parameter count and FLOPs are significantly less than BC-YOLO. Compared to YOLOv5s, YOLOv7-tiny, and GC-YOLO, LiteYOLO-ID shows a decrease in recall rate and mAP (0.5–0.95), but has a significant advantage in precision and mAP (0.5). Additionally, its parameter count is reduced by 47.13%, 38.37%, and 61.52%, while FLOPs are reduced by 44.94%, 34.09%, and 70%, respectively. Compared to ML-YOLOv5, Faster R-CNN-tiny, and Improved YOLOv5, our model is only slightly lower in the mAP (0.5–0.95) metric but outperforms these three models in all other metrics. Compared to YOLOv5-GCDNet, LiteYOLO-ID surpasses YOLOv5-GCDNet in all detection performance metrics, given the same parameter count. These results demonstrate that in practical applications, our model can maintain high detection accuracy while running at lower computational costs, making it highly suitable for deployment on devices with limited computational resources.

Fig. 8 presents the schematic of the detection results by LiteYOLO-ID. We employed techniques of increasing brightness and image flipping to enhance the test images. Fig. 8(a)–(c), respectively, show the detection results of insulator flashovers, which are primarily caused by the electric current breaking through the insulation layer and forming conductive pathways along the surface of the insulator. The visual characteristics are mainly manifested as ablation or melting marks on the insulator surface that differ from the surrounding areas. Meanwhile, Fig. 8(d)–(f) represents the damaged conditions of insulators, typically caused by physical impacts, material aging, or temperature changes, with the main visual features being obvious cracks or complete fractures on the insulators. Through Fig. 8, we can clearly see that our model is capable of effectively detecting different types of insulator defects.

As shown in Fig. 9, we compared the LiteYOLO-ID with the original YOLOv5s model. By comparing images (a) and (c), we can observe the phenomenon of redundant detection in the YOLOv5s model; whereas the comparison between images (b) and (d) reveals instances of missed detections in YOLOv5s. From Fig. 9, it is evident that the visualization effect of LiteYOLO-ID is overall superior to that of the original model.

4) Comparative Experiment of Pascal VOC Datasets: To further validate the performance of our model, we conducted experiments on the Pascal VOC dataset [58], [59], which consists of 16 551 training images and 4952 validation images, covering 20 distinct detection categories. On the Pascal VOC

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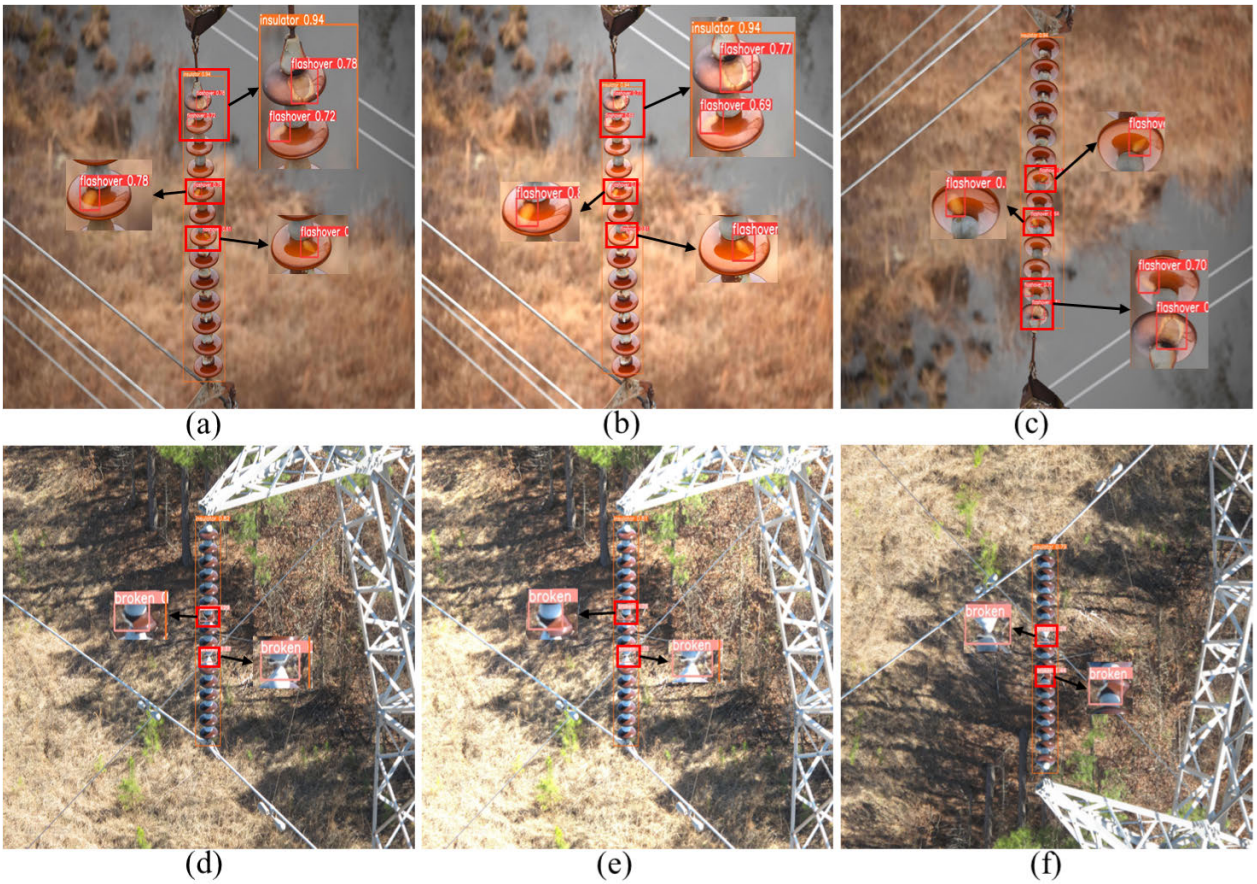


Fig. 8. LiteYOLO-ID detection result diagram. (a) Represents the detection results of flashover. (b) Represents the detection results of flashover after brightness enhancement. (c) Represents the detection results of flashover after flipping. (d) Represents the detection results of damage. (e) Represents the detection results of damage after brightness enhancement. (f) Represents the detection results of damage after flipping.

TABLE V
PASCAL VOC DATASET EXPERIMENTAL RESULTS

Dataset	Model	Param.(M)	FLOPs(G)	P	R	mAP(0.5)	mAP(0.5-0.95)
Pascal VOC	YOLOv3-tiny [22]	8.71	13	61.1%	53%	55.9%	26.2%
	YOLOv4-tiny [23]	5.92	16.2	76.96%	33.79%	52.5%	21.6%
	YOLOv5n [24]	1.79	4.2	73.3%	68.5%	73.1%	46.3%
	Efficientdet_d0 [60]	3.84	4.71	62.34%	13.46%	30.16%	13.8%
	YOLOv7-tiny [27]	6.07	13.3	75.6%	76.1%	79.7%	54.5%
	YOLOv8s [28]	11.13	28.5	83.8%	75.6%	83.5%	63.7%
	YOLOv5s [24]	7.06	15.9	79.1%	72.5%	79.1%	54.1%
	LiteYOLO-ID(ours)	3.76	8.9	78.3%	72.7%	78.5%	53.9%

dataset, all models were experimented with using nearly identical hyperparameters, and the detection results are presented in Table V.

From Table V, it can be observed that compared to YOLOv3-tiny, YOLOv4-tiny, YOLOv5n, and Efficientdet_d0 [60], LiteYOLO-ID demonstrates superior performance across various detection metrics. In comparison to YOLOv5s, our model has successfully reduced the number of parameters and FLOPs by 46.82% and 44.03%, respectively, without significant loss in detection accuracy. Although our model shows slightly less effectiveness in terms of detection accuracy compared to YOLOv7-tiny and YOLOv8s, it has achieved a reduction in the number of parameters and FLOPs by 38.06% and 66.22%, and 33.08% and 68.77%, respectively. These results indicate that our model not only maintains a balance between accuracy and parameter reduction but also exhibits good generalizability.

5) Comparative Experiment of SFID Datasets: To verify the generalization ability of LiteYOLO-ID across different insulator defect datasets, we conducted validation on the publicly available large-scale insulator defect dataset SFID [32]. This dataset comprises 10 975 training images and 2743 test images, encompassing two detection categories: normal and defective. The experimental results on the SFID dataset are presented in Table VI. Some of the data are derived from the original research by Zhang et al. [32] on the SFID dataset.

As shown in Table VI, our model not only performs excellently in terms of accuracy, but more importantly, it has a lower number of parameters, making it more suitable for embedded deployment.

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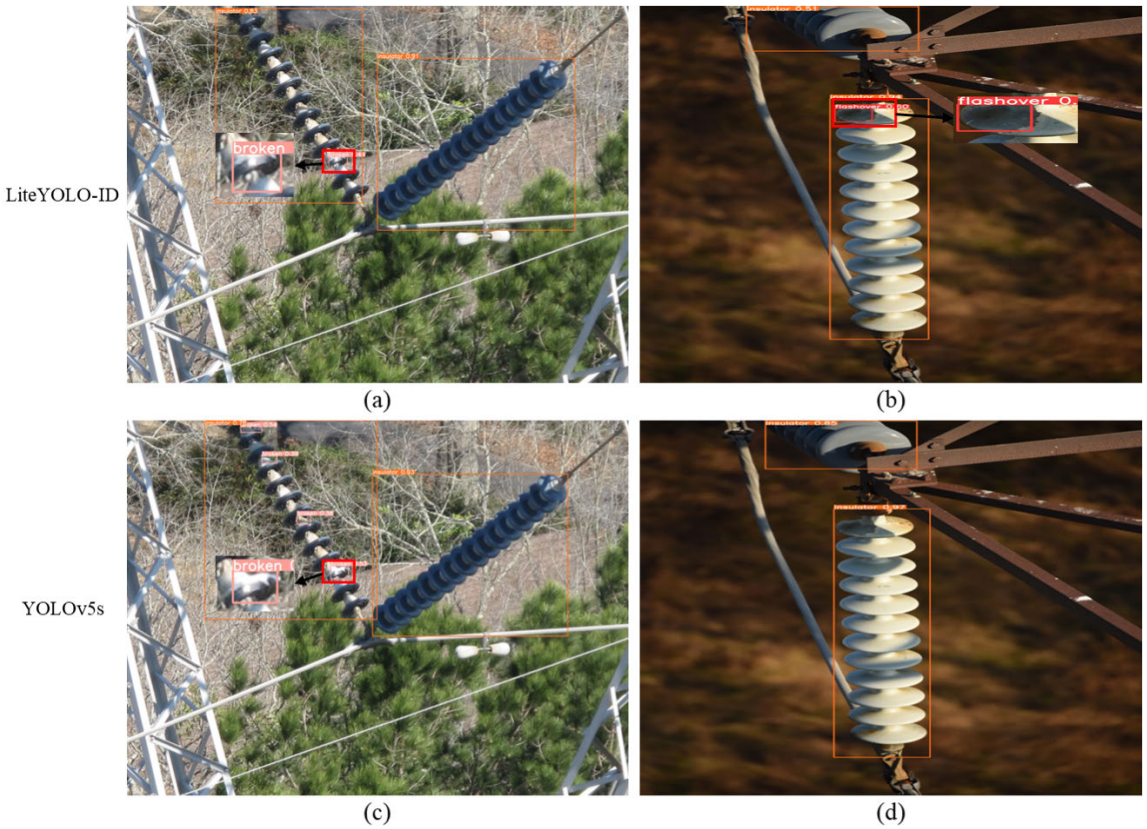


Fig. 9. Comparison chart of detection results. (a) and (c)Images of insulator damage defects. (b) and (d) Images of insulator flashover defects.

TABLE VI
SFID DATASET EXPERIMENTAL RESULTS

Dataset	Model	mAP(0.5)	mAP(0.5-0.95)
SFID	FINet [32]	99.5%	88.3%
	Faster RCNN [14]	98.4%	80.1%
	Mask RCNN [15]	98.3%	82%
	YOLOX [25]	99.4%	86%
	Swin-Transformer [61]	99%	86.4%
	YOLOv5s [24]	99.3%	87%
	LiteYOLO-ID(ours)	99.4%	84.9%

TABLE VII
EXPERIMENTAL RESULTS OF MODEL INFERENCE SPEED COMPARISON ON EDGE DEVICES

Equipment	Model	FPS
Jstson TX2 NX	YOLOv4-tiny [23]	7.38
	YOLOX-tiny [25]	6.21
	YOLOv5-GCDNet [36]	4.47
	Improved YOLOv4-tiny [37]	5.95
	ML-YOLOv5 [38]	18.83
	Faster R-CNN-tiny [19]	8.67
	GC-YOLO [39]	6.64
	BC-YOLO [40]	13.25
	Improved YOLOv5 [41]	11.67
	YOLOv8s [28]	15.83
	YOLOv5s [24]	17.48
	LiteYOLO-ID(ours)	20.2

C. Testing on Edge Computing Devices

Considering that LiteYOLO-ID is designed specifically for embedded deployment applications, we optimized the model using TensorRT and conducted actual inference speed tests on the Jetson TX2 NX, with Fig. 10 showing the actual detection output. In the same testing environment, we utilized the same dataset for evaluation and conducted comparative experiments with mainstream YOLO series models and recently released insulator detection models. The experimental results are shown in Table VII. According to Table VII, our model demonstrated excellent performance, with a frame rate of up to 20.2, significantly surpassing the other models. This result indicates that LiteYOLO-ID is highly suitable for applications requiring real-time performance.

D. Ablation Study

Through ablation experiments, we assessed the impact of specific components of the network model on overall performance. The results of these ablation experiments are presented in Table VIII.

From Table VIII, it is evident that when using only EGC-CSPGhostNet as the backbone network, all detection metrics improve compared to the original model. When EGC-PANet is used solely as the neck network, compared to the original model, the number of parameters and FLOPs decrease by 23.65% and 17.09%, respectively, while the accuracy and mAP (0.5) increase by 1.3% and 0.1%, respectively. However, the recall rate and mAP (0.5–0.95) decrease by 1.6% and 0.3%, respectively. When both EGC-CSPGhostNet and EGC-PANet are employed as the backbone and neck networks, respectively, there is a reduction of 47.13% in parameters and 44.94% in FLOPs compared to the original model. In terms of

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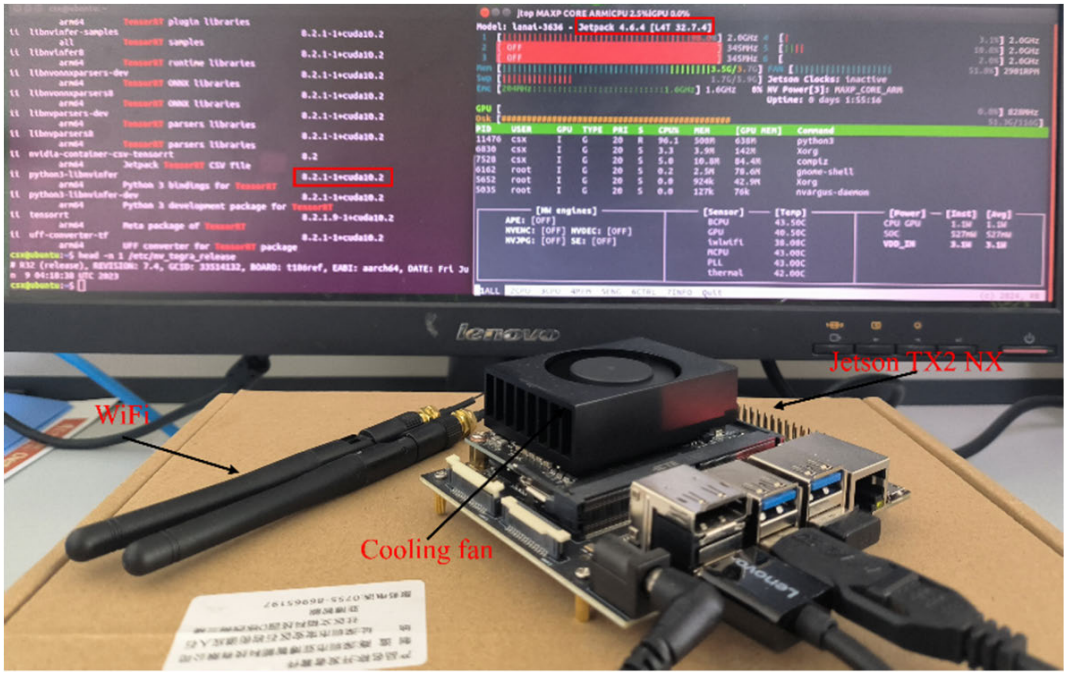


Fig. 10. Actual detection output on Jetson TX2 NX.

TABLE VIII
ABLATION EXPERIMENT RESULTS

Dataset	+EGC-CSPGhostNet	+EGC-PANet	+YOLO Head	Param.(M)	FLOPs(G)	P	R	mAP(0.5)	mAP(0.5-0.95)
IDID-Plus	-	-	√	7.02	15.8	75.5%	60.9%	64.1%	39.4%
	√	-	√	5.33	11.3	75.8%	61.7%	64.4%	37.4%
	-	√	√	5.36	13.1	76.8%	59.3%	64.2%	39.1%
	√	√	√	3.71	8.7	83%	59.5%	65.1%	37.5%

accuracy, mAP (0.5) increases by 1%, and precision increases by 7.5%, while the recall rate and mAP (0.5–0.9) only decrease by 1.4% and 1.9%, respectively.

V. CONCLUSION

To address the challenge of deploying insulator defect detection models on edge devices, this article introduces a lightweight insulator defect detection model, LiteYOLO-ID. To suit real-world application scenarios, we constructed the IDID-Plus dataset. The experimental results show that on the IDID-Plus dataset, LiteYOLO-ID, compared to the original YOLOv5s detection model, not only increases the mAP (0.5) by 1% but also reduces the parameter count by 47.13% and the FLOPs by 44.94%. This validates the effectiveness of LiteYOLO-ID in terms of detection performance. Moreover, on the public datasets PascalVOC and SFID, LiteYOLO-ID achieves a good balance between detection accuracy and model parameter count, further proving the algorithm's robust generalization capability. We optimized the improved model with TensorRT and deployed it on Jetson TX2 NX, verifying the model's practical inference capability.

Compared to existing models, our proposed method has lower computational costs and can be easily deployed on edge devices for real-time insulator defect detection. In future studies, we plan to adopt cloud computing and Internet of Things technologies to integrate image capture, object detection, and data transmission functions, achieving real-time detection of insulator defects on drone platforms. Additionally, we will explore applying our model to various practical scenarios such as road maintenance and bridge inspection. We also intend to utilize image preprocessing techniques and collect more high-quality datasets of insulator defects to enhance the model's applicability.

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